

DESIGN AND ANALYSIS OF ALGORITHM (CS-327)

LECTURE – 6

MASTER THEOREM METHOD

Another method to solve recurrence relation, but in this method we can't solve every recurrence relation we only solve specific type of recurrence relations.

Example :-

$$T(n) = aT(n/b) + f(n)$$

Example :-

$$T(n) = 4T(n/2) + n.$$

Solution:-

$$T(n) = 4T(n/2) + n \text{ ----- (i)}$$

from general equation

$$T(n) = aT(n/b) + f(n) \text{ ----- (ii)}$$

where

$$a=4$$

$$b=2$$

$$f(n) = n^1$$

$$\theta(n^1)$$

$$d=1$$

because $4 > 2$

$$a > b^d$$

so by comparing it matches with condition (iii) in formulae list.

Basic Formulae to remember

$$\text{if } f(n) = \theta(n^d) \quad d \geq 0$$

$$(i) \quad T(n) = \theta(n^d) \quad \text{if } a < b^d$$

$$(ii) \quad T(n) = \theta(n^d \log n) \quad \text{if } a = b^d$$

$$(iii) \quad T(n) = \theta(n^{\log_b a}) \quad \text{if } a > b^d$$

$$\begin{aligned}
T(n) &= \theta(n^{\log_b a}) \\
&= \theta(n^{\log_2 4}) \\
&= \theta(n^{\log_2 2 \times 2}) \\
&= \theta(n^{2 \log_2 2})
\end{aligned}$$

$$T(n) = \theta(n^2)$$

Answer = $\theta(n^2)$ time complexity of this recurrence relation.

Example :-

$$T(n) = 3T(n/2) + n^2.$$

Solution:-

$$T(n) = 3T(n/2) + n^2 \text{ -----(i)}$$

from general equation

$$T(n) = aT(n/b) + f(n) \text{ -----(ii)}$$

$$a=3, b=2, d=2.$$

$$a < b^d$$

$$3 < 2^2$$

$$3 < 4$$

so by comparing it matches with condition (i) in formulae list.

$$T(n) = \theta(n^d)$$

$$T(n) = \theta(n^2)$$

Answer = $\theta(n^2)$ time complexity of this recurrence relation.

Practice Problems :- (DIY)

$$- T(n) = 2T(n/2) + n$$

$$- T(n) = 4T(n/3) + n$$

$$- T(n) = 4T(n/2) + n^2$$

More Formulae to Remember:-

$$T(n) = aT(n/b) + \theta(n^k \log^p n)$$

(1) if $a > b^k$; $T(n) = \theta(n^{\log_b a})$

(2) if $a = b^k$; so

(i) if $P < -1$; $T(n) = \theta(n^{\log_b a})$

(ii) if $p = -1$; $T(n) = \theta(n^{\log_b a} \cdot \log^2 n)$

(iii) if $P > -1$; $T(n) = \theta(n^{\log_b a} \cdot \log^{p+1} n)$

(3) if $a < b^k$; so

(i) if $p < 0$; $T(n) = \theta(n^k)$

(ii) if $p \geq 0$; $T(n) = \theta(n^k \log^p n)$

Example :-

$$T(n) = 2T(n/2) + n \log n$$

by comparing with general form

$$T(n) = aT(n/b) + \theta(n^k \log^p n)$$

$$a = 2, b = 2, k = 1, P = 1.$$

Since it is matching with condition 2 which is $a = b^k$ now we will check the condition of P , so we have $P=1$ ($p > -1$ in formula).

$$T(n) = \theta(n^{\log_b a} \cdot \log^{p+1} n)$$

$$= \theta(n^{\log^2 2} \cdot \log^{1+1} n)$$

$$= \theta(n^1 \log^2 n)$$

$T(n) = \theta(n \log^2 n)$ is the time complexity of this recurrence relation.

Example :-

$$T(n) = 3T(n/3) + n/2$$

by comparing with general form

$$T(n) = aT(n/b) + \theta(n^k \log^p n)$$

$$a=3, b=3, k=1, p=0.$$

Since it is matching with condition 2 which is $a = b^k$ now we will check the condition of P , so we have $P=0$ ($p > -1$ in formula).

$$T(n) = \theta(n^{\log_b a} \cdot \log^{p+1} n)$$

$$= \theta(n^{\log_3 3} \cdot \log^{0+1} n)$$

$$= \theta(n^1 \cdot \log^1 n)$$

$T(n) = \theta(n \log n)$ is the time complexity of this recurrence relation.

Practice Problems :- (DIY)

- $T(n) = \sqrt{2}T(n/2) + \log n$
- $T(n) = 8T(n/4) - n^2 \log n$